

Rushil Karani

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EDUCATION

McMaster University (B. Eng)

Sep. 2023 – Apr. 2027

Honours Software Engineering (3.9 GPA), Dean's Honour Roll (2024, 2025)

Hamilton, ON

Relevant Coursework: DSA, Databases, OOP (Java), Software Design, Concurrency, Linear Optimization.

EXPERIENCE

Full-Stack Developer (Incoming), Co-op

Sep. 2026 – Present

Royal Bank of Canada

Toronto, ON

- Incoming Full-Stack Developer in the marketing technology team @RBC.

Software Developer, Co-op

May 2026 – Aug. 2026

Bell Canada

Mississauga, ON

- Architected a full-stack **Next.js/TypeScript** app for managing IVR broadcast messages, cutting update turnaround to **under 60s**. Processed **35,000 lines** (8MB) in under **20 seconds** using a bulk CSV pipeline.
- Minimized **Azure Cosmos DB** load and cost by serving all high-frequency API reads from an in-memory cache (**60s TTL**). Secured external IVR-client APIs with **SHA-256-hashed** keys rate-limited to **10,000 req/min**.
- Deployed to Azure Container Apps with auto-scaling (**1→3 replicas**) and achieved zero-downtime releases.
- Built IRIS, an intern onboarding tool with a built-in **LLM assistant**. Featured low-latency search using in-browser **SQLite**. Won **2nd place** at Bell's intern hackathon and drew deployment interest from senior leadership.

Automation Analyst, Co-op

Jan. 2026 – Apr. 2026

Royal Bank of Canada

Toronto, ON

- Automated PDF report validation by building a **Python/React** tool, reducing manual verification time by **96%**. Improved throughput by **10x** using parallel processing and caching across multiple report formats.
- Improved API reliability by developing an automated health-check system to scan **22 endpoints** each day and send Slack alerts on failures. Detected **6 critical issues** and reduced manual monitoring effort by **50%**.
- Debugged and validated a regression test suite through **GitHub Actions CI pipelines**, ensuring full test coverage and catching **3 critical defects** before merging to production.

PROJECTS

PrepPal (Won 3rd Place @ MEC) | | TypeScript, Google Cloud Vision

Nov. 2025

- Integrated Google Cloud Vision API to identify essential evacuation items from user-provided images.
- Achieved **85%** recognition accuracy on common household items by refining and tuning the dataset.
- Developed an emergency chatbot that generates first-aid guidance and provides appropriate online resources.

Health Connect | | JavaScript, Node.js, React, Express, MongoDB

Jan. 2025

- Developed a healthcare web application, to reduce wait times in 5 walk-in clinics by **up to 20%**.
- Used the **MERN stack** for full stack app development, increasing development speed by **5 hours**.
- Developed an AI chatbot using **Cohere's API** and resolved 3 complex issues in the chatbot and UI.

Maze Runner | | Java, JUnit 5, Maven

Apr. 2025

- Built a maze solver app in Java using Maven for build automation, **reducing build time by 30%**.
- Created 10+ unit tests using **JUnit 5** to handle edge cases, increasing code **reliability by 25%**.
- Used OOP principles/patterns (MVC, Observer, Polymorphism) to improve **maintainability by 40%**.

Equi-Hire | | Python, Google Cloud Vision, Java, JavaScript, React, SQL

Feb. 2025

- Built an AI resume screening tool that reduced hiring biases **by 30%** and sped up hiring process **by 10%**.
- Troubleshooted 3 complex issues, overcoming challenges like **API integration** and **Git merge** conflicts.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++.

Frameworks & Tools: React, Node.js, Express, Spring Boot, Git, GitHub Actions, CI/CD, Linux.

Technologies: REST APIs, Distributed Systems, Caching, Parallel Processing, Databases (MongoDB, MySQL).